BIOINFORMATICS BOOT CAMP

Assessment 03 – Sequence Analysis and GC/AT Content Analysis

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Assignment No: 03

Q1. Sequence Analysis

Sequence analysis is the study of DNA, RNA and protein sequences to determine their structure, function and evolutionary relationships. It is important for gene discovery, disease diagnosis, evolutionary studies and drug development.

Q2. GC and AT Content

GC content is the percentage of guanine and cytosine while AT content is the percentage of adenine and thymine in DNA. These values help determine DNA stability and genome characteristics.

Q3. Scientific Problem

Accession Number: NM_000546.6. Example calculations: Total Length = 2500 nucleotides, A=650, T=620, G=620, C=610, GC%=49.2, AT%=50.8.

Q4. Data Interpretation

Sequence B (68% GC) is expected to be more thermally stable because GC pairs form three hydrogen bonds compared to two in AT pairs.

Q5. Practical Activity

NCBI nucleotide databases provide sequence information used for biological analysis. FASTA format stores nucleotide sequences in a standardized way.

Q6. Critical Thinking

Yes. Two DNA sequences can have the same length but different GC contents due to different nucleotide compositions, resulting in different thermal stability.